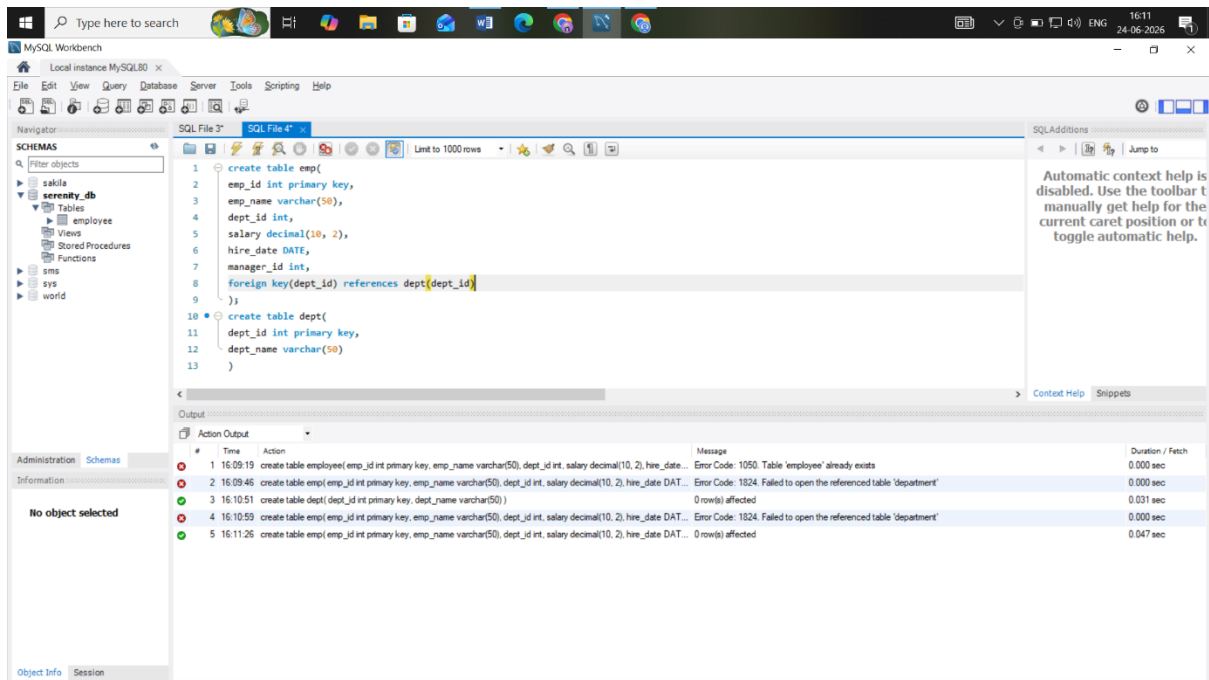


Task- SQL queries

1.Create an Employee table and department table.

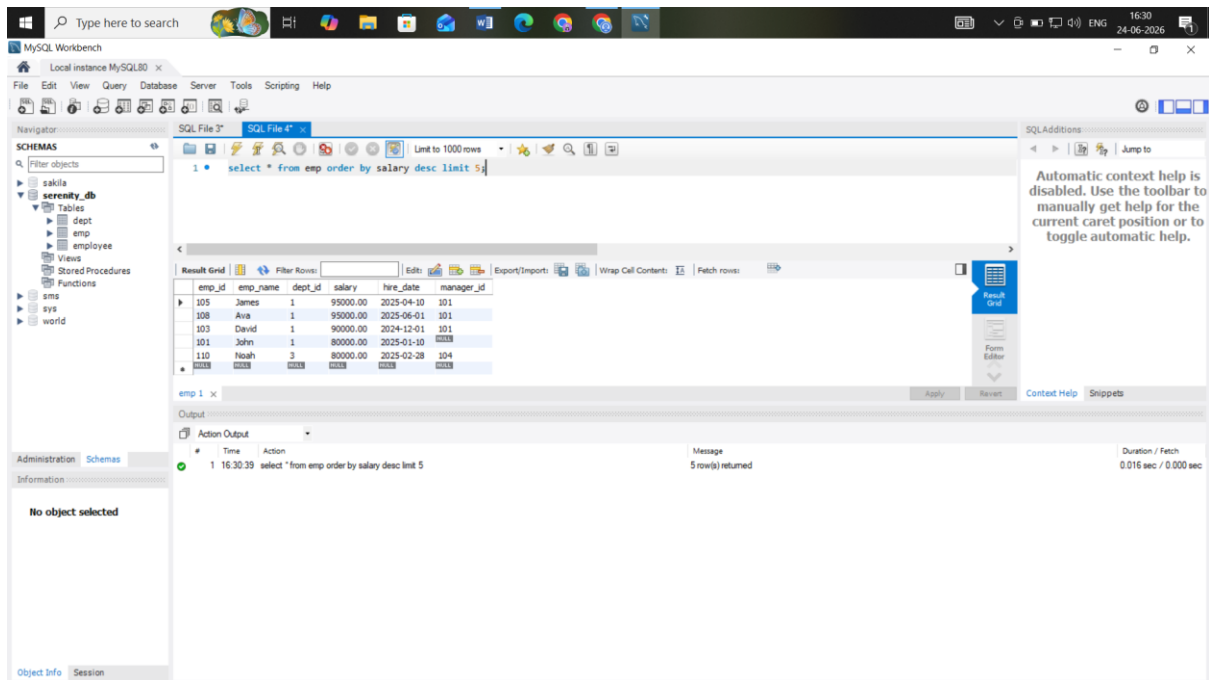
```
create table dept(  
dept_id int primary key,  
dept_name varchar(50)  
)
```

```
create table dept(  
dept_id int primary key,  
dept_name varchar(50)  
)
```



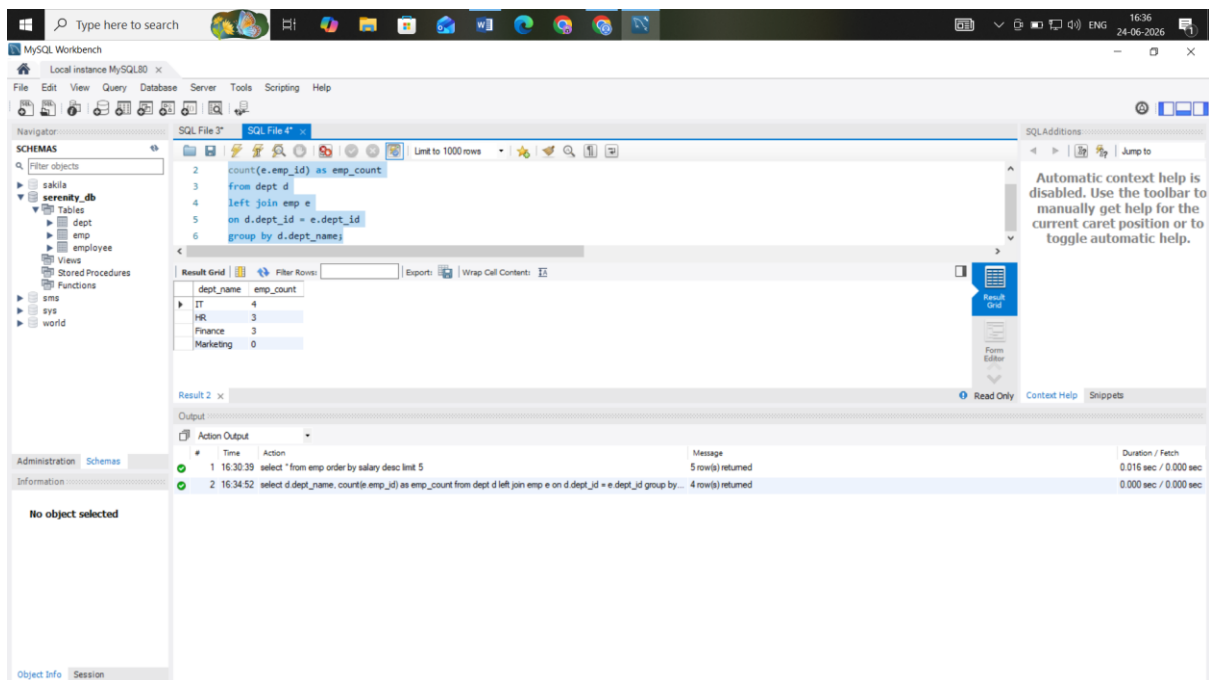
2.Top 5 Highest Salary

```
select * from emp order by salary desc limit 5;
```



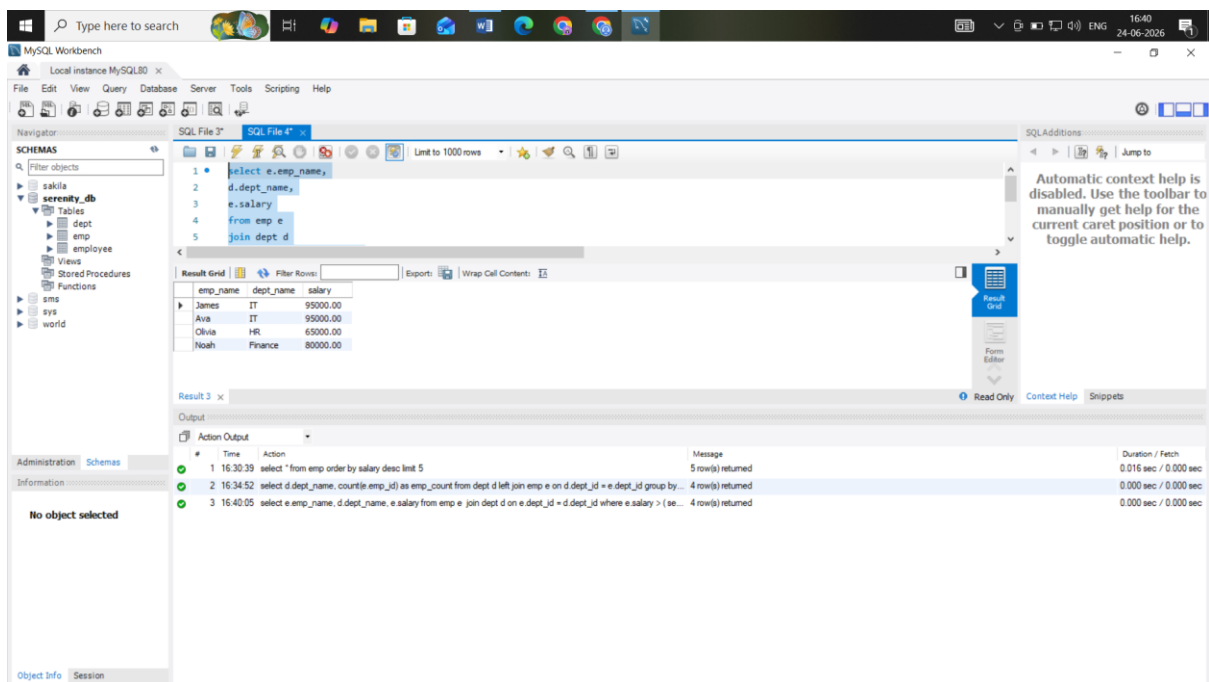
3.Department wise employee count

```
select d.dept_name,
count(e.emp_id) as emp_count
from dept d
left join emp e
on d.dept_id = e.dept_id
group by d.dept_name;
```



4. Employees whose salary > department avg salary

```
select e.emp_name,  
       d.dept_name,  
       e.salary  
from emp e  
join dept d  
on e.dept_id = d.dept_id  
where e.salary > (  
select avg(salary)  
from emp  
where dept_id = e.dept_id  
);
```



5. Inner join

```
select e.emp_id,  
       e.emp_name,  
       d.dept_name  
from emp e  
inner join dept d on e.dept_id = d.dept_id;
```

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

- sakila
- serenity_db
 - Tables
 - dept
 - emp
 - employee
 - Views
 - Stored Procedures
 - Functions
- sys
- world

SQL File 3*

```

1 select e.emp_id,
2 e.emp_name,
3 d.dept_name
4 from emp e
5 inner join dept d on e.dept_id = d.dept_id;

```

Result Grid

emp_id	emp_name	dept_name
101	John	IT
103	David	IT
105	James	IT
108	Ava	IT
102	Emma	HR
106	Olivia	HR
109	Mia	HR

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	16:30:39	select * from emp order by salary desc limit 5	5 row(s) returned	0.016 sec / 0.000 sec
2	16:34:52	select d.dept_name, count(e.emp_id) as emp_count from dept d left join emp e on d.dept_id = e.dept_id group by...	4 row(s) returned	0.000 sec / 0.000 sec
3	16:40:05	select e.emp_name, d.dept_name, e.salary from emp e join dept d on e.dept_id = d.dept_id where e.salary > (se...	4 row(s) returned	0.000 sec / 0.000 sec
4	16:43:24	select e.emp_id, e.emp_name, d.dept_name from emp e inner join dept d on e.dept_id = d.dept_id LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Administration Schemas

Information

No object selected

Object Info Session

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

6.left join

select

d.dept_name,

e.emp_name

from dept d

left join emp e on d.dept_id = e.dept_id;

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

- sakila
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 - emp
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 - Views
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 - Functions
- sys
- world

SQL File 3*

```

1 select
2 d.dept_name,
3 e.emp_name
4 from dept d
5 left join emp e on d.dept_id = e.dept_id;

```

Result Grid

dept_name	emp_name
IT	John
IT	David
IT	James
IT	Ava
HR	Emma
HR	Olivia
HR	Mia
Finance	Sophia
Finance	William
Finance	Noah
Marketing	

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	16:46:58	select d.dept_name, e.emp_name from dept d left join emp e on d.dept_id = e.dept_id LIMIT 0, 1000	11 row(s) returned	0.000 sec / 0.000 sec

Administration Schemas

Information

No object selected

Object Info Session

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

7.Group by with having

```
select d.dept_name,  
count(*) as total_employees  
from emp e  
join dept d  
on e.dept_id = d.dept_id  
group by d.dept_name  
having count(*) > 2;
```

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
1 select d.dept_name,  
2 count(*) as total_employees  
3 from emp e  
4 join dept d  
5 on e.dept_id = d.dept_id  
6 group by d.dept_name  
7 having count(*) > 2;
```

The Results tab shows the following data:

dept_name	total_employees
IT	4
HR	3
Finance	3

The Action Output tab shows the execution progress:

#	Time	Action	Message	Duration / Fetch
1	16.46.58	select d.dept_name, e.emp_name from dept d left join emp e on d.dept_id = e.dept_id LIMIT 0, 1000	11 row(s) returned	0.000 sec / 0.000 sec
2	16.50.11	select d.dept_name, count(*) as total_employees from emp e join dept d on e.dept_id = d.dept_id group by d.dept_name having count(*) > 2	3 row(s) returned	0.015 sec / 0.000 sec

8.Employee hired in last month

```
select * from emp where hire_date >= date_sub(curdate(), interval 15 month);
```

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
1 where hire_date >= date_sub(curdate(), interval 15 month);
```

The Results tab shows the following data:

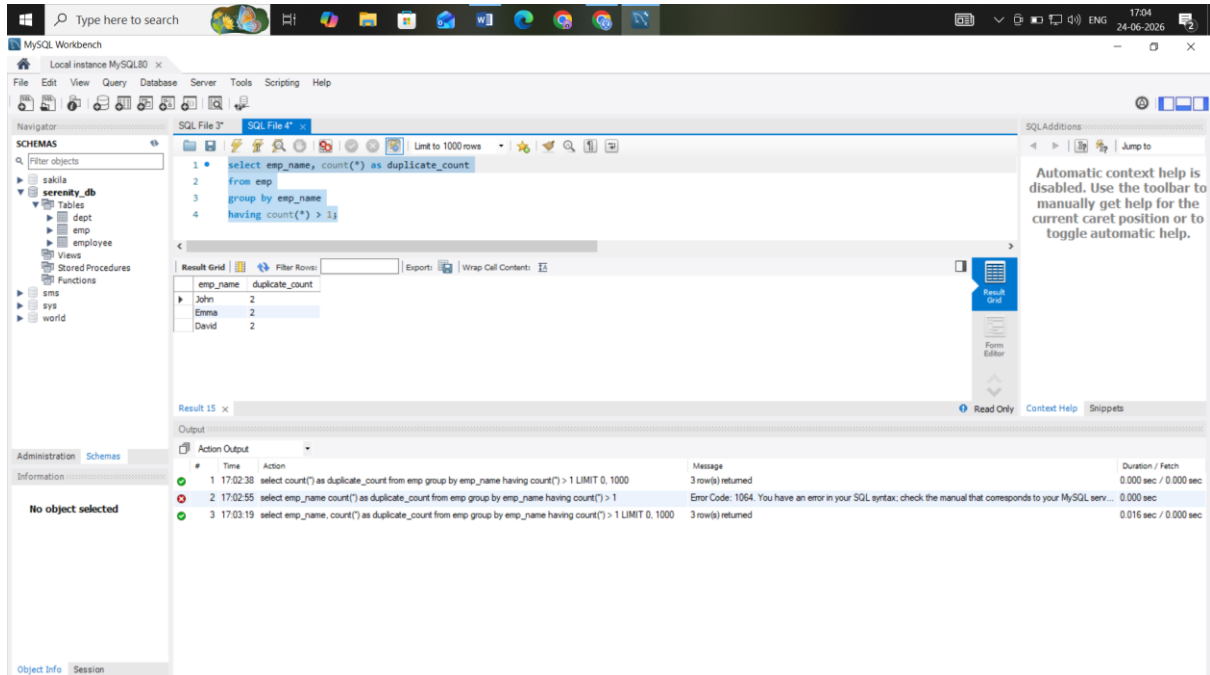
emp_id	emp_name	dept_id	salary	hire_date	manager_id
105	James	1	95000.00	2025-04-30	101
106	Olivia	2	65000.00	2025-05-05	102
108	Ava	1	95000.00	2025-06-01	101

The Action Output tab shows the execution progress:

#	Time	Action	Message	Duration / Fetch
1	16.46.58	select d.dept_name, e.emp_name from dept d left join emp e on d.dept_id = e.dept_id LIMIT 0, 1000	11 row(s) returned	0.000 sec / 0.000 sec
2	16.50.11	select d.dept_name, count(*) as total_employees from emp e join dept d on e.dept_id = d.dept_id group by d.dept_name having count(*) > 2	3 row(s) returned	0.015 sec / 0.000 sec
3	16.53.02	select * from emp where hire_date >= date_sub(curdate(), interval 15 month) LIMIT 0, 1000	0 row(s) returned	0.031 sec / 0.000 sec
4	16.53.20	select * from emp where hire_date >= date_sub(curdate(), interval 10 month) LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
5	16.53.28	select * from emp where hire_date >= date_sub(curdate(), interval 10 month) LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
6	16.53.41	select * from emp where hire_date >= date_sub(curdate(), interval 15 month) LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

9. Duplicate Records

```
select emp_name, count(*) as duplicate_count  
from emp  
group by emp_name  
having count(*) > 1;
```



10. Remove the duplicate records

```
DELETE e1  
FROM emp e1  
JOIN emp e2  
ON e1.emp_name = e2.emp_name  
AND e1.dept_id = e2.dept_id  
AND e1.salary = e2.salary  
AND e1.emp_id > e2.emp_id;
```

